Networking Academy

Packet Tracer - Design and Implement a VLSM Addressing Scheme

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Police	G0/0 (PD-2)	192.168.203.97	255.255.255.224	N/A
	G0/1 (PD-1)	192.168.203.129	255.255.255.240	N/A
	S0/0/0 (Schools)	192.168.203.[illegible]	255.255.255.252	N/A
Schools	G0/0 (PS-101)	192.168.203.65	255.255.255.[illegible]224	N/A
	G0/1 (PS-115)	192.168.203.1	255.255.255.[illegible]192	N/A
	S0/0/0 (Police)	192.168.203.[illegible]	255.255.255.252	N/A
PD-1	VLAN 1	192.168.203.130	255.255.255.240	192.168.203.129
PD-2	VLAN 1	192.168.203.98	255.255.255.224	192.168.203.97
PS-101	VLAN 1	192.168.203.66	255.255.255.224	192.168.203.65
PS-115	VLAN 1	192.168.203.2	255.255.255.192	192.168.203.1
PD-1-11	NIC	192.168.203.142	255.255.255.240	192.168.203.129
PD-2-23	NIC	192.168.203.126	255.255.255.224	192.168.203.97
PS-101-57	NIC	192.168.203.94	255.255.255.224	192.168.203.65
PS-115-12	NIC	192.168.203.62	255.255.255.192	192.168.203.1

Objectives

In this lab you will design a VLSM addressing scheme given a network address and host requirements. You will configure addressing on routers, switches, and network hosts.

- Design a VLSM IP addressing scheme given requirements.
- Configure addressing on network devices and hosts.
- Verify IP connectivity.
- Troubleshoot connectivity issues as required.

Background / Scenario

You have been asked to design, implement, and test an addressing scheme for a customer. The customer has given you the network address that is suitable for the network, the topology, and the host requirements. You will implement and test your design.

Instructions

You have been given the network address 192.168.203.0/24 by your customer. The host address requirements are:

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Requirements

Host Requirements:

LAN / WAN	Number of Addresses Required
/28 PD-1 LAN	6
/27 PD-2 LAN	19
/27 PS-101 LAN	21
/26 PS-115 LAN	32
/30 Police - Schools WAN	2

Design Requirements

- Create the addressing design. Follow guidelines provided in the curriculum regarding the order of the subnets.
- The subnets should be contiguous. There should be no unused address space between subnets.
- Provide the most efficient subnet possible for the point-to-point link between the routers.
- Document your design in a table such as the one below.

Subnet Description	Number of Hosts Needed	Network Address/CIDR	First Usable Host Address	Broadcast Address
PS-115 LAN	32	192.168.203.0/26	192.168.203.1	192.168.203.63
PS-101 LAN	21	192.168.203.64/27	192.168.203.65	192.168.203.95
PD-2 LAN	19	192.168.203.96/27	192.168.203.97	192.168.203.127
PD-1 LAN	6	192.168.203.128/29	192.168.203.129	192.168.203.135
Police - Schools WAN	2	192.168.203.144/30	192.168.203.145	192.168.203.147

Configuration Requirements

Note: You will configure addressing on **all** devices and hosts in the network.

- Assign the first usable IP addresses in the appropriate subnets to Police for the two LAN links and the WAN link.
- Assign the first usable IP addresses in the appropriate subnets to Schools for the two LANs links. Assign the last usable IP address for the WAN link.
- Assign the second usable IP addresses in the appropriate subnets to the switches.
- The switch management interface should be reachable from hosts on all the LANs.
- Assign the last usable IP addresses in the appropriate subnets to the hosts.

If the addressing design and implementation are correct, all hosts and devices should be reachable over the network.

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